

- 4 (a) The product of the force and the perpendicular distance of its line of action from the pivot. [1]

- (b) (i) Component of the weight perpendicular to rod = $5.2 \sin 56^\circ$

Taking moments about A,

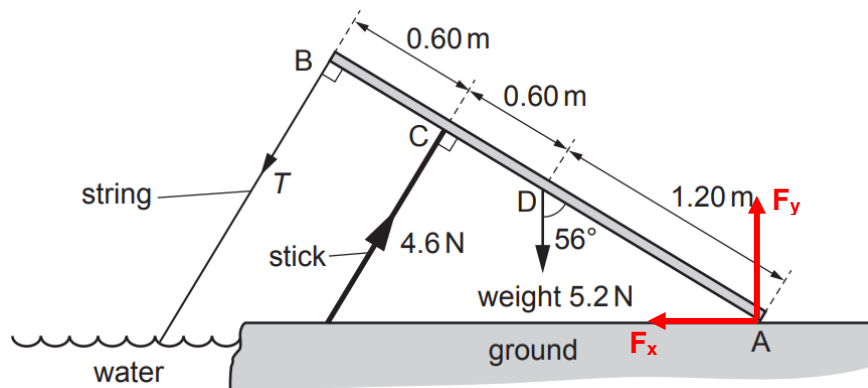
$$(T)(2.40) + (5.2 \sin 56^\circ)(1.20) = (4.6)(1.80)$$

$$T = 1.29 \text{ N}$$

[1]

[1]

(ii)



Horizontally, $F_x + T \cos 56^\circ = 4.6 \cos 56^\circ$
 $F_x = 1.85 \text{ N}$

Vertically, $F_y + 4.6 \sin 56^\circ = T \sin 56^\circ + 5.2$
 $F_y = 2.46 \text{ N}$

[1 for both F_x and F_y ECF from (b)(i)]

Magnitude of $F = 3.1 \text{ N}$

[1]

Angle with respect to rod = $\tan^{-1} (2.460 / 1.848) - 34^\circ = 19^\circ$

[1]

5 (a) (i) Since resistance is the ratio of potential difference to current and from the graph